

1 two dimensional

1.1 Corresponding deterministic version

We use average population density of all patches (including infected and susceptible ones) to replace stochastic events and keep other mechanisms unchanged.

Order of events is **population growth** $\rightarrow$ **host movement and infection between patches** $\rightarrow$ **fizzle** $\rightarrow$ **burnout**.

This is equivalent to **burnout** $\rightarrow$ **population growth** $\rightarrow$ **host movement and infection between patches** $\rightarrow$ **fizzle**. (And this will make equations more concise.)

1.1.1 Burnout

$$f_b : \binom{p}{N} \mapsto \binom{P_{\text{nb}}p}{(1 - pzD)N}$$

where $P_{\text{nb}} = 1 - P_{\text{b}}$ is the probability of not burning out conditioning on not fizzling. And number of infected rats is $np_0zN_0 = npzN$, z is the final size, n is total number of patches, and we use N_0 and p_0 to denote initial N and p value at the beginning of the iteration.

1.1.2 Population growth

$$f_g : \binom{p}{N} \mapsto \binom{p}{N(1 + r(1 - \frac{N}{K}))}$$

1.1.3 Movement and infection

$$f_i : \binom{p}{N} \mapsto \binom{p + (1 - e^{-\alpha cnpzN_0})(1 - p)}{N}$$

1.1.4 Fizzle

$$f_f : \binom{p}{N} \mapsto \binom{P_{\text{nf}}p}{N}$$

where $P_{\text{nf}} = 1 - P_{\text{f}}$ is the probability of not fizzling.

1.1.5 Overall map

The overall map is $f = f_f \circ f_i \circ f_g \circ f_b$

$$f : \binom{p}{N} \mapsto \binom{(P_{\text{nb}}p + (1 - P_{\text{nb}}p)(1 - e^{-\alpha cnpzN}))P_{\text{nf}}}{N(1 - pzD)(1 + r(1 - \frac{(1 - pzD)N}{K}))}$$

where P_{nb} is evaluated at $N_0 = N$.

The Jacobian matrix at DFE ($p = 0, N = K$) is

$$J_f(0, K) = \begin{pmatrix} P_{\text{nf}}P_{\text{nb}} + \alpha cznK & 0 \\ -(1 - r)KzD & 1 - r \end{pmatrix}$$

So the persisence threshold is

$$L_0 = P_{\text{nf}}P_{\text{nb}} + \alpha cznK > 1$$

1.2 Comparison with previous deterministic model

In our previous two dimensional deterministic model our threshold is

$$L_0 = P_1 + B > 1$$

where $P_1 = P_{\text{nf}}P_{\text{nb}}$ is persistence probability (neither fizzling nor burning out).

B and $\alpha cznK$ are both constant, so the two deterministic versions are equivalent. However the range of $\mathcal{R}_0$ we want was very narrow in the deterministic version, indicating that difference in population density is crucial in explaining virulence evolution.

2 three dimensional

2.1 Corresponding deterministic version

2.1.1 Burnout

$$f_b(p, N_I, N_S) = \left(\frac{\begin{matrix} P_{\text{nb}}p \\ (1 - zD)N_I \end{matrix}}{\frac{(1 - zD)N_IP_{\text{b}}p + N_S(1 - p)}{P_{\text{b}}p + 1 - p}} \right)$$

2.1.2 Population growth

$$f_g(p, N_I, N_S) = \left(\frac{\begin{matrix} p \\ N_I + rN_I \left(1 - \frac{N_I}{K}\right) \end{matrix}}{\begin{matrix} N_S + rN_S \left(1 - \frac{N_S}{K}\right) \end{matrix}} \right)$$

2.1.3 Movement

$$f_m(p, N_I, N_S) = \left(\frac{\begin{matrix} p \\ N_I + c(1 - p)(N_S - N_I) \end{matrix}}{N_S - cp(N_S - N_I)} \right)$$

2.1.4 Infection

$$f_i(p, N_I, N_S) = \left(\frac{\begin{matrix} p + (1 - e^{-\alpha cnpzN_{I_0}})(1 - p) \\ pN_I + (1 - e^{-\alpha cnpzN_{I_0}})(1 - p)N_S \end{matrix}}{\frac{p + (1 - e^{-\alpha cnpzN_{I_0}})(1 - p)}{N_S}} \right)$$

2.1.5 Fizzle

$$f_f(p, N_I, N_S) = \left(\frac{\begin{matrix} P_{\text{nf}}p \\ N_I \end{matrix}}{\frac{(1 - p)N_S + P_{\text{f}}pN_I}{1 - p + P_{\text{f}}p}} \right)$$

2.1.6 Overall map

$$f = f_f \circ f_i \circ f_m \circ f_g \circ f_b$$